Received: 20 January 2023 | Revised: 24 March 2023 | Accepted: 25 March 2023
DOI: 10.1111/jerd.13044

REVIEW ARTICLE

WILEY

An overview of the digital occlusion technologies: Intraoral scanners, jaw tracking systems, and computerized occlusal analysis devices

Marta Revilla-León DDS, MSD, PhD[1,2,3] | Dean E. Kois DMD, MSD[4,5] | Jonathan M. Zeitler[6] | Wael Att DDS, Dr Med Dent, PhD[7] | John C. Kois DMD, MSD[4,5,8]

[1]Department of Restorative Dentistry, School of Dentistry, University of Washington, Seattle, Washington, USA
[2]Research and Digital Dentistry, Kois Center, Seattle, Washington, USA
[3]Department of Prosthodontics, School of Dental Medicine, Tufts University, Boston, Massachusetts, USA
[4]Kois Center, Seattle, Washington, USA
[5]Private Practice, Seattle, Washington, USA
[6]IT, Kois Center, Seattle, Washington, USA
[7]Department of Prosthodontics, School of Dental Medicine, Tuff University, Boston, Massachusetts, USA
[8]Department of Restorative Dentistry, School of Dental Medicine, University of Washington, Seattle, Washington, USA

Correspondence
Marta Revilla-León, 1001 Fairview Ave N # 2200, Seattle, WA 98109, USA.
Email: marta.revilla.leon@gmail.com

Abstract

Objectives: Within the development of digital technologies, dental professionals aim to integrate virtual diagnostic articulated casts obtained by using intraoral scanners (IOSs), the mandibular motion of the patient recorded by using an optical jaw tracking system, and the information provided by computerized occlusal analysis systems. This article describes the various digital technologies available for obtaining the digital occlusion of a patient and outlines its challenges and limitations.

Overview: The factors that influence the accuracy of the maxillomandibular relationship of diagnostic casts obtained by using IOSs are reviewed, as well as the occurrence of occlusal collisions or mesh interpenetrations. Different jaw tracking systems with varying digital technologies including ultrasonic systems, photometric devices, and artificial intelligence algorithms are reviewed. Computerized occlusal analysis systems for detecting occlusal contacts in a time sequential manner with the pressure distribution on the occlusal surfaces are reviewed.

Conclusions: Digital technologies provide powerful diagnostic and design tools for prosthodontic care. However, the accuracy of these digital technologies for acquiring and analyzing the static and dynamic occlusion need to be further analyzed.

Clinical Significance: Efficiently implementing digital technologies into dental practice requires an understanding of the limitations and state of current development of the digital acquisition methods for digitizing the static and dynamic occlusion of a patient by using IOSs, digital jaw trackers, and computerized occlusal analysis devices.

KEYWORDS
accuracy, computerized occlusal analysis devices, digital impressions, digital occlusion, digital scans, intraoral scanners, jaw tracking systems

1 | INTRODUCTION

Dental occlusion, or the static relationship between the incising or masticating surfaces of the maxillary or mandibular teeth or tooth analogues,[1] is a critical component for diagnosing and executing a prosthodontic treatment plan, using either conventional or digital methods. The mandibular motion of the patient has been recorded and studied since the early 19th century.[2–10] The purpose of these recordings is to integrate the mandibular motion of the patient into diagnostic and treatment planning procedures, as well as to be used for the designing and fabricating of interim and definitive prostheses.

Different technologies have been described for tracking and recording the mandibular motion of the patient, including mechanical devices such as the Gnathic replicator,[11] the Kinesiograph,[12] or the

Axiograph,[13] photographic methods,[3,4] roentgenography,[14] electronic and telemetric techniques,[15,16] magnetometry,[17] and opto-electronic systems.[18,19] Similarly, dental articulators have been developed to replicate the mandibular motion of the patient for designing and fabricating dental restorations.[20–25] Within the incorporation of digital technologies in dentistry, additional armamentarium such as ultrasonic systems,[26–28] photometric devices,[29–33] and artificial intelligence (AI)[34–37] algorithms have been described for tracking and recording the mandibular motion of the patient.[38] Additionally, in some of these systems, the recorded mandibular motion can be integrated into the computer-aided design (CAD) software programs for designing dental prostheses.[26–33]

The growing use of intraoral scanners (IOSs) for different dental interventions,[39] including acquiring virtual articulated diagnostic and definitive casts, provides a powerful digital alternative for digitizing and virtually analyzing the static and dynamic dental occlusion of a patient, although the accuracy and reliability of these digital methods for occlusal analysis is still unclear.[40–45] Additionally, computerized occlusal analysis devices offer a digital alternative to quantitatively assess and visualize the pressure and timing of occlusal contacts.[41,46–49]

Classical gnathology assumed that dental occlusion was a mechanical constant and reproducible.[46,47] However, studies published in the dental literature have revealed occlusal variations can result from recordings made at different times of day, stress, proprioception, body position, and temporomandibular joint dysfunction.[48–58] Conventional methods, including different occlusal recording materials and the use of foils, also have limitations in a clinical application. A gold standard does not exist.[59–62]

To realize the potential of these digital methodologies in restorative and prosthodontics, a systematic categorization and characterization of the development, performance, and limitations of these systems is needed. The objectives of this review article were to provide an overview of the current systems available to capture the digital occlusion by using IOSs, optical jaw tracking systems, and computerized occlusal analysis systems while reviewing the performance of these systems.

2 | INTRAORAL SCANNERS

IOSs systems are capable of recording the maxillomandibular relationship by integrating virtual occlusal records that are used to align the previously acquired virtual diagnostic casts (Figure 1).[63,64] Additionally, IOSs systems can also provide information regarding contacts in static and dynamic occlusion. The operator and patient factors that can influence the scanning accuracy of IOSs,[65,66] as well as the variables that can affect the accuracy of the maxillomandibular relationship of the virtual casts obtained by using IOSs, have been analyzed in the dental literature.[63,64]

Operator skills and decision-making influence the accuracy of IOSs, including the selection of the IOS technology and system used, scanning head size, calibration, scanning distance and angulation, exposure of the IOS to ambient temperature changes,[67] ambient humidity, ambient lighting conditions,[68,69] operator experience, scanning pattern, extension of the scan, and the use of cutting off, rescanning, and overlapping procedures. Patient factors are defined as the intraoral conditions of the patient being scanned that influence the scanning accuracy of IOSs. These patient factors are tooth type, presence of interdental spaces, arch width, palate characteristics, wetness, existing restorations, characteristics of the surface being digitized, edentulous areas, interimplant distance, the position, angulation, and depth of existing implants, and implant scan body design.[66]

Overall, independent of the scanning technology and IOSs system, IOSs provide a reliable digital impression alternative for acquiring virtual diagnostic casts with similar accuracy to conventional impression methods.[70,71] Clinical studies have evaluated the accuracy of IOSs for acquiring complete-arch intraoral digital scans, reporting a

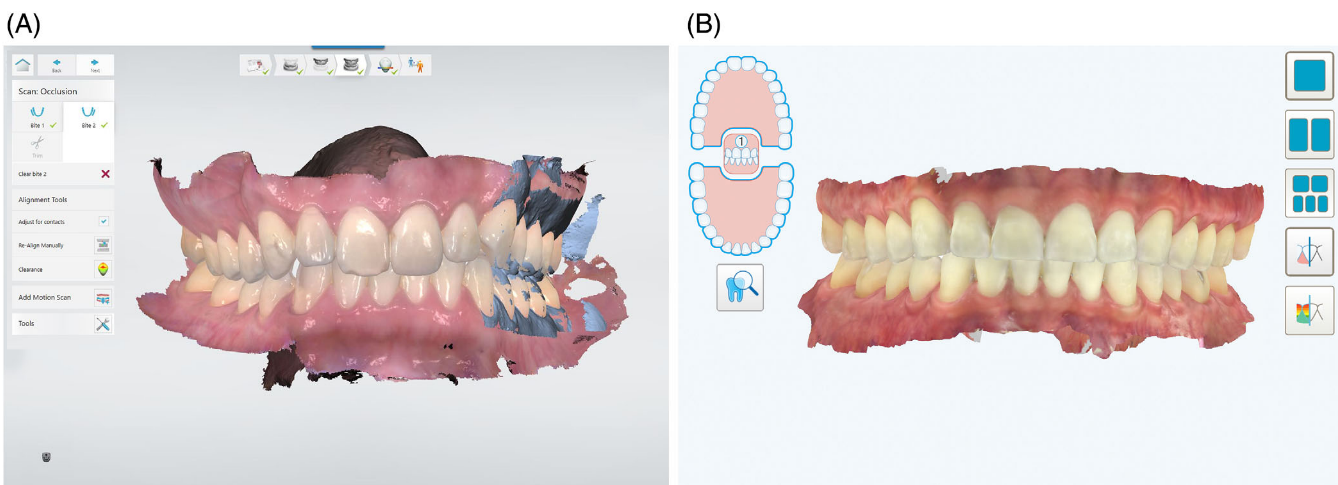

FIGURE 1 Static maxillomandibular relationship recorded by using an intraoral scanner. (A) (Trios 4, wireless, from 3Shape A/S. (B) Itero Element 5D Plus from Align Technologies.

trueness mean value ranging from 73 to 433 μm and a precision mean value ranging from 80 to 199 μm.[65,66,72–74]

Previous systematic reviews have assessed the accuracy of the virtual static articulation obtained by using IOSs.[63,64] Both reviews agreed that IOS systems can obtain similar static articulations as conventional impression methods.[64,65] However, one claimed that this static virtual articulation is as accurate as conventional techniques only in the presence of completely dentate arches, stable occlusal contacts, a single prepared tooth, or arches involving a single missing posterior tooth.[64] Additionally, different variables that can influence the accuracy of the static virtual articulation acquired by using IOSs have been identified including the articulation technique, characteristics of the virtual occlusal records (number, dimension, and location), length of the digitized arches, presence of edentulous areas (position and size), and IOS technology.[64,75–77]

An in-vitro study assessed the influence of the number of teeth (2, 3, or 4 teeth) included in the bilateral virtual occlusal records on the accuracy of the maxillomandibular relationship acquired by using an IOS.[75] The results revealed that bilateral occlusal records that involved 4 teeth obtained higher maxillomandibular accuracy than bilateral occlusal records that included less than 4 teeth.[75] Therefore when possible, it might be recommended to include at least 4 teeth on the bilateral occlusal records captured by using IOSs.

Dental literature has reported the plasticity of the periodontal ligament necessary to maintain the homeostasis within the periodontium.[38] The fibroblasts of the periodontal ligament are a significant factor in regulating the adaptation of the periodontium to extrinsic and intrinsic stimuli. The resiliency, or plasticity, of the periodontal ligament allows ∼100 μm of tooth micromovement .[38] Occlusal collisions or mesh interpenetrations can be identified when virtual articulated casts penetrate one into each other (Figure 2).[36,77–79] Occlusal collisions are caused by the tooth location discrepancy resulting from the periodontal ligament plasticity between the recording of intraoral digital scans of the maxillary and mandibular arches and the virtual occlusal records.[64] The intraoral digital scans of both arches are obtained under unloaded conditions, while the virtual occlusal records are usually acquired in maximum intercuspation position (MIP).[64] As a consequence, mesh interpenetrations between the maxillary and mandibular arches may occur. Mesh interpenetrations could also be influenced by the different intraoral scanning technologies and IOS postprocessing procedures; however, dental literature

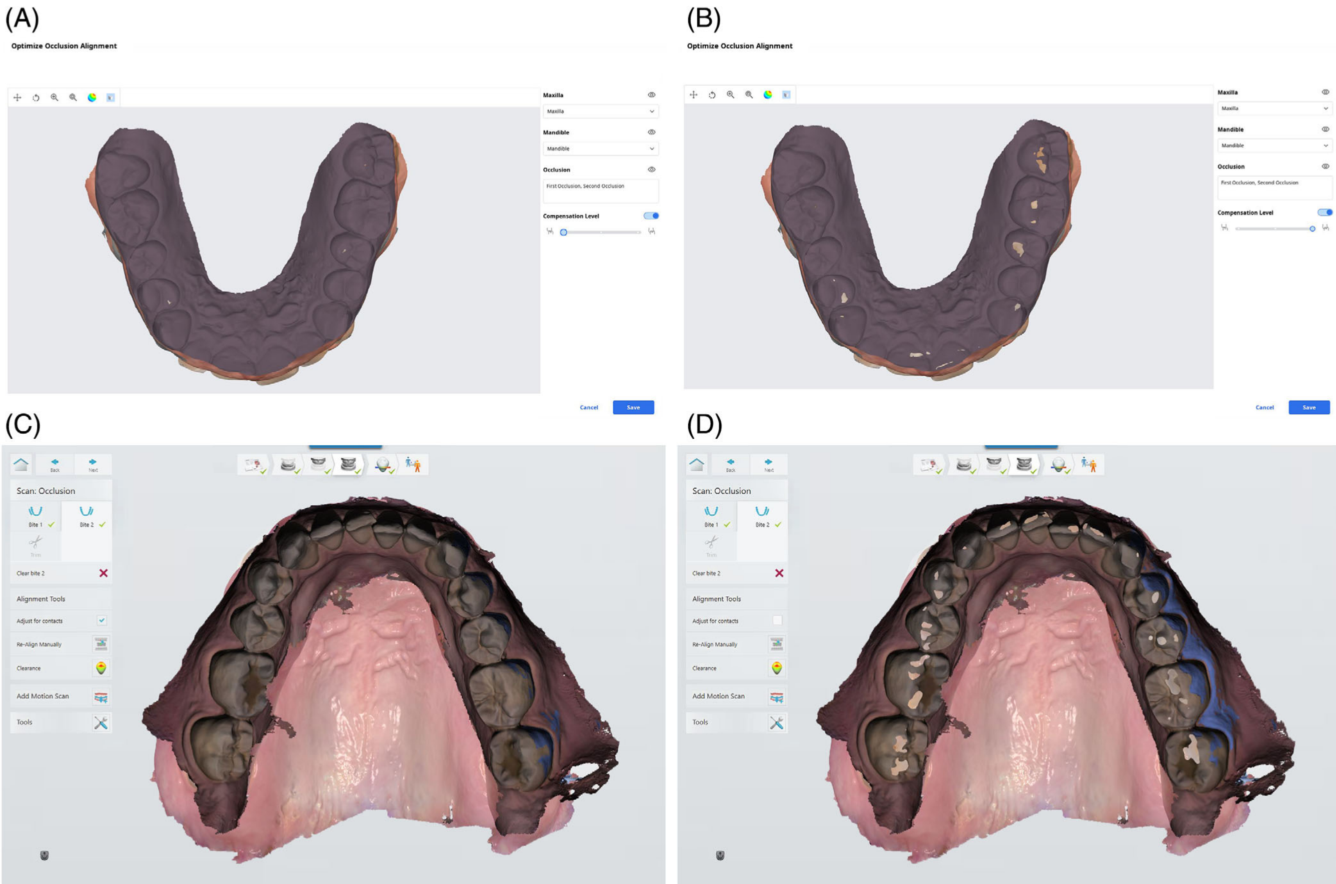

FIGURE 2 Virtual collision in articulated virtual diagnostic casts. (A) Articulated casts without occlusal collisions captured by using an IOS (i700, wireless; Medit). (B) Articulated casts without occlusal collisions captured by using an IOS (i700, wireless; Medit). (C) Articulated casts without occlusal collisions captured by using an IOS (Trios 4, wireless; 3Shape A/S). (D, A) Articulated casts with occlusal collisions captured by using an IOS (Trios 4, wireless; 3Shape A/S).

analyzing the influence of the technology and postprocessing procedures of the IOSs on the presence of occlusal collisions is scarce.

Mesh interpenetrations produced by any digital registration method, if left uncorrected, could result in undercontoured or noncontacting restored teeth.[79] Some IOSs software programs allow the selection of different levels of mesh interpenetration corrections; however, the effect of such corrections on the accuracy of the articulated virtual diagnostic casts remains unknown. Additionally, some CAD software programs detect such mesh interpenetrations and suggest correction by either virtually removing volume (the interpenetration areas) on the occlusal surfaces on the virtual casts or opening the maxillomandibular relationship until there are interpenetration areas. However, the influence of such CAD corrections on the maxillomandibular relationship accuracy or designs of the dental restorations is unknown.

Revilla-León et al.[76] evaluated the influence of the absence (MIP) or presence of interocclusal space (0°, 1°, 2°, 3°, or 4° of incisal opening in a semi-adjustable articulator) on the maxillomandibular relationship accuracy captured by using an IOS. The results showed that the greater the amount of interocclusal space available, the higher the accuracy of the maxillomandibular relationship measurement.[76] The authors postulated that less overlapping between the maxillary and mandibular dentition may favor more accurate alignment procedures.[76] Additionally, the absence of interocclusal contacts when having interocclusal space and, therefore, lack of occlusal collusions on the articulated virtual casts, may be contributed to the results reported. This study identified the need for additional in vitro and clinical studies to assess the influence of the interocclusal space on the accuracy of the maxillomandibular relationship, as well as its relationship with maxillary and mandibular dentition overlapping and occlusal collisions.

In the opinion of the authors, while digital technologies may improve clinical efficiency and provide more powerful diagnostic tools, dental professionals face the challenge of uncertainty. Digital technologies, or in this case IOSs, offer the capability of a fast data acquisition method that allows, when combined with the analysis competency of the varying CAD software programs, a higher amount of information to be collected and analyzed when compared with conventional methods. However, new digital technology-related variables should be understood and considered when applying the technology clinically, as well as knowing the accuracy of the digital measurement methods assessed. Digital technologies have the potential to analyze data in a way that has not been possible with conventional methods and may facilitate a better understanding of dental occlusion. The development of evidence-based criteria is fundamental for supporting the implementation of these IOSs in private practice. Further laboratory and clinical studies analyzing the accuracy of articulated virtual casts obtained by using IOSs are needed.

Conventional methods are currently considered the gold standard and, therefore, used as a control to compare the outcome of IOSs. However, traditional impression methods also incorporate distortion and may not represent the most accurate technique to acquire patient information. In vitro studies facilitate the evaluation of the gross accuracy of IOSs, as well as the assessment of different variables that can influence the IOSs outcome. However, in a clinical environment, the scanning conditions and IOSs performance may not be reflected in laboratory settings. The development of alternative evaluation methods within the development and implementation of digital technologies may provide a new gold standard.

AI algorithms have been developed for restorative and prosthodontic applications.[80–82] AI models, such as smart stitching or AI cleaning, have also been implemented in IOS software programs aiming to facilitate the scanning procedure; however, the impact of such AI applications is unclear. Similarly, the proprietary knowledge of IOS manufacturers regarding their software stitching, maxillomandibular alignments, and occlusal collision adjustments are a challenge to better understanding and making comparisons among the different IOS systems.

3 | DIGITAL JAW TRACKING SYSTEMS

Digital jaw systems that are available both commercially and noncommercially have been described.[26–34] The main digital technologies include ultrasonic systems,[26–28] infrared optical camera systems, photometric devices,[29–33] IOSs, and AI[34–37,77] algorithms. Jaw trackers can be classified based on the functional capabilities that the system offers: diagnostic (data acquisition and data analysis) systems or diagnostic and designing systems (Table 1). Diagnostic jaw tracking systems allow the tracking and recording of the mandibular motion of the patient and provide a posterior analysis within the system's software program, but the recording cannot be exported. Systems with diagnostic and designing capabilities (CAD integrable) are diagnostic jaw tracking systems that allow the exporting of the patient's mandibular motion to be integrated into CAD software programs.

Ultrasonic systems employ ultrasonic collectors attached to a head coordinate frame and emitters located into the tracker attached to the mandibular dentition (JMA System; Zebris Medical GmbH).[26] The system determines the mandibular relative position by calculating the time for ultrasonic pulses sent from emitters to reach collectors. The 3D mandibular motion is then recreated by aligning a triangular plane between the condyle points and the base position of the system. This triangular plane is then used to set the target position of the mandible. Ultrasound-based devices are sensitive to environmental noise and temperature which may affect the positional accuracy up to 0.1 mm after calibration.[34]

Infrared optical camera systems consist of a facebow (receiver and control unit) and the lower jaw sensor (transmitter unit) (JMA-Optic System; Zebris Medical GmbH). The lower jaw sensor is powered by a battery button cell. It contains 12 LEDs that are activated during measurement by an infrared signal from the facebow. During a measurement, the movements of the sensor, which is attached to the mandibular teeth, are recorded by the cameras integrated into the facebow (Figure 3A). The position of the luminous points of the LEDs are determined by the two cameras in the facebow and sent to the computer via USB connection. The computer uses this data to

TABLE 1 Description of the main commercially available digital jaw tracking systems.

Jaw tracker classification	Manufacturer	Jaw tracker	Technology	Tracking markers	CAD integration
Diagnostic (data acquisition and analysis)	Bioresearch	JT-3D	Magnetometry	Head frame, mandibular tracker	NA
	Myotronics	K7x Jaw Tracking CMS	Magnetometry	Head frame, mandibular tracker	
Diagnostic (data acquisition and analysis) and designing capabilities (CAD integrable)	BiONiC	Bionic Jaw Motion	Photometric system	Maxillary and mandibular tracker	XML file
	IGNIDent	DMD (dental motion decoder) system	Magnetometry	Maxillary and mandibular tracker	
	Itaka	Cyclops System	Photometric system	Head frame, mandibular tracker	
	Kavo	Arcus Digmall	Ultrasound system	Head frame, mandibular tracker, virtual earbow	
	Modjaw	4D Motion Capture System	Photometric system	Head frame, mandibular tracker, wand	
	Orangedental	Freecorder Bluefox 2.0	Photometric system	Head frame, mandibular tracker	
	Planmeca	4D Jaw Motion	Photometric system with CBCT	Head frame, mandibular tracker	
	SDiMatrix	Proaxis	Photometric system	Head frame, maxillary and mandibular tracker	
	Zebris Medical GmbH	JMA	Ultrasound system	Head frame, mandibular tracker, virtual earbow	
		JMA-Optic	Infrared optical system	Head frame, mandibular tracker, virtual earbow	

Abbreviations: CAD, computer-aided design; CBCT, cone beam computed tomography; NA, not applicable; XML, extensible markup language file.

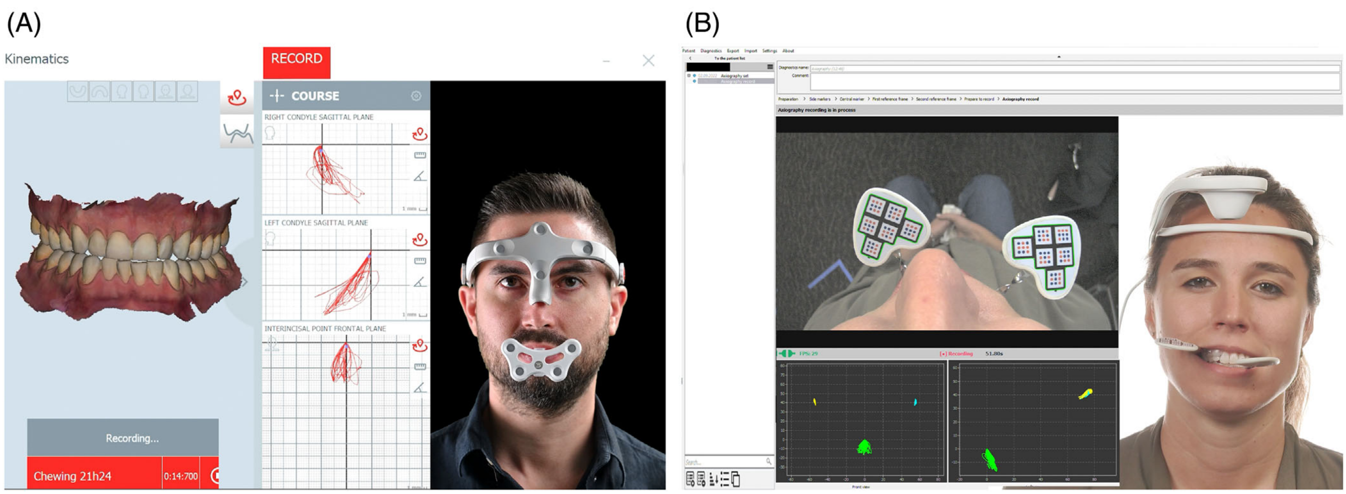

FIGURE 3 Examples of optical mandibular jaw trackers. (A) 4D Motion Capture System from Modjaw. (B) Proaxis system from SDIMatrix.

determine the position of the lower jaw sensor relative to the face-bow. The application software is used to store and analyze the measurement data.

Photometric devices use one or more cameras to track the position of markers fixed to the mandible, with or without additional markers positioned on the face of the patient (Figure 3B).[29–33] Video-based technologies for the mandibular motion tracking have been reported as highly dependent on the size of the markers and the frame rate of video capture, with smaller markers performing poorly when compared with high frame-per-second recording devices.[83]

Some IOSs provide the capability to record the mandibular motion of the patient; however, the accuracy of these IOS recordings is unknown. Last, AI algorithms have been described that aim to automate the mandibular motion recording; however, its accuracy is unclear.[34–37,77]

The accuracy of these jaw tracking systems remains unclear.[31,84–86] A recent systematic review assessed the accuracy of the systems that digitally analyze the mandibular motion, as well as reviewing the physiological and device-dependent factors that influence the accuracy of these systems.[84] However in this systematic review of

30 articles, only half of the studies were rated as having a moderate to high certainty of evidence.[84] The authors concluded that realistic variations in jaw tracking device accuracy ranged from 50 to 330 μm across the digital systems with very low interoperator reliability for motion tracking from photographs.[84] Additionally, mandibular and condylar growth, kinematic dysfunction of the neuromuscular system, shortened dental arches, previous orthodontic treatment, variations in habitual head posture, temporomandibular joint disorders, fricative phonetics, and to a limited extent, parafunctional habits and unbalanced occlusal contact were identified as confounding variables that shaped jaw movement trajectories, but were not highly dependent on age, gender, or diet.[84]

Li et al.[31] evaluated the accuracy of an experimental jaw tracking system. A maxillary and mandibular complete denture mounted on a semi-adjustable articulator were used to perform small opening and excursive movements.[31] The motion of the casts was recorded by the experimental jaw tracker and the 3D displacements were compared. The discrepancy measured in protrusion was 131 ± 39 μm, in left laterotrusion was 133 ± 44 μm, in right laterotrusion was 120 ± 51 μm, and in opening was 112 ± 52 μm.[31] However, this study was completed in a laboratory setting and the jaw tracker is not commercially available.

In a clinical study, Revilla-León et al.[32] compared the accuracy of the maxillomandibular relationship captured with a Kois deprogrammer by using four IOSs (TRIOS 4 from 3Shape A/S, iTero Element 5D from Align Technologies, i700 from Medit, and Primescan from Dentsply Sirona) and an optical jaw tracking system (4D Motion Capture System; Modjaw). The maxillomandibular relationship trueness values from high to low reported were iTero (0.14 ± 0.09 mm), followed by the Modjaw (0.20 ± 0.04 mm) and the TRIOS4 (0.22 ± 0.09 mm) groups. However, the iTero, Modjaw, and TRIOS4 groups were not significantly different from each other. The i700 group obtained the lowest trueness and precision values (0.40 ± 0.22 mm) of all groups tested, followed by the Primescan group (0.26 ± 0.13 mm); however, the i700 and Primescan groups had significantly lower trueness and precision than only the iTero group.[32]

In a posterior clinical study completed by the same research team,[86] authors analyzed the influence of the imported IOS system into the accuracy of the maxillomandibular relationship captured with a Kois deprogrammer by using an optical jaw tracking system (4D Motion Capture System; Modjaw). Except for the i700 IOS system, the optical jaw tracking system tested improved the trueness value of the maxillomandibular relationship recorded when compared with the corresponding IOS.[86]

Additional studies are needed to further assess the accuracy of these ultrasonic, optical camera systems, photometric devices, IOSs, and AI algorithms for tracking and recording the mandibular motion of a patient, as well as the variables that influence its accuracy. Furthermore, studies are needed to assess the accuracy (trueness and precision), efficiency, and clinical impact when these recorded mandibular motions are used for designing interim and definitive restorations or occlusal devices. Digital technologies also require the overcoming of a learning curve.

4 | COMPUTERIZED OCCLUSAL ANALYSIS SYSTEMS

The sensitivity and reliability of the conventional occlusal contact indicators are influenced by the thickness, strength, and elasticity of the material, saliva absorption, and interpretation of the clinician.[57,59,87,88] Different computerized occlusal analysis systems have been described for detecting occlusal contacts in a time sequential manner with the pressure distribution on the occlusal surfaces (Figure 4, Table 2).[41,46–49,89–93] Additionally, IOSs can also detect occlusal contact areas (OCAs) (Figure 5).[41]

The accuracy of these occlusal analysis systems has been assessed in the dental literature.[41,46–49,89–93] Computerized occlusal analysis systems provide a reliable method for measuring the OCA at maximum bite force; however, conventional occlusal registration methods seem to provide the highest reliability and validity technique for measuring OCA.[41,48,91–93]

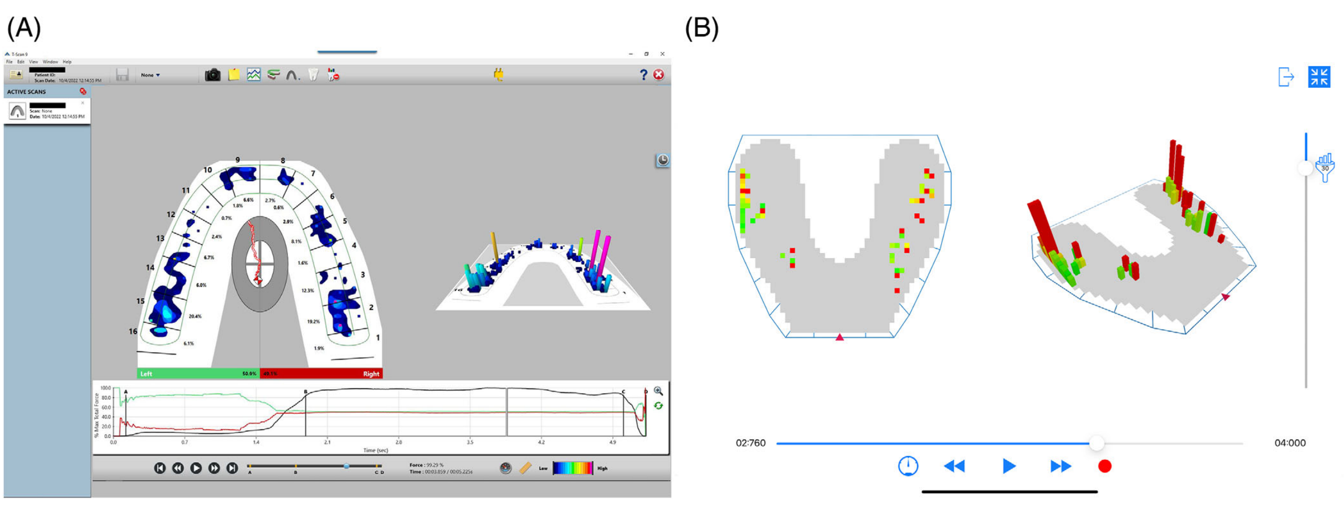

FIGURE 4 Computerized occlusal analysis systems. (A) T-scan from Tekscan. (B) OccluSense from Bausch.

TABLE 2 Description of the main commercially available computerized occlusal analysis systems.

Manufacturer	Occlusal analysis system	Functionality	Occlusal sensor	Occlusal sensor thickness
Bausch	OccluSense	Static and dynamic occlusal contacts (pressure and timing)	OccluSense Electronic Pressure Sensor (Large and extra-large sizes)	60 μm
Dmtec	Accura	Static and dynamic occlusal contacts (pressure and timing)	Occlusal sensor (S and M sizes)	160 μm
Tekscan	T-scan Novus	Static occlusal contacts (pressure and timing) Total bite force (N)	T-scan sensor (S and L sizes)	100 μm

Abbreviations: L, large; M, medium; N, Newton; S, small.

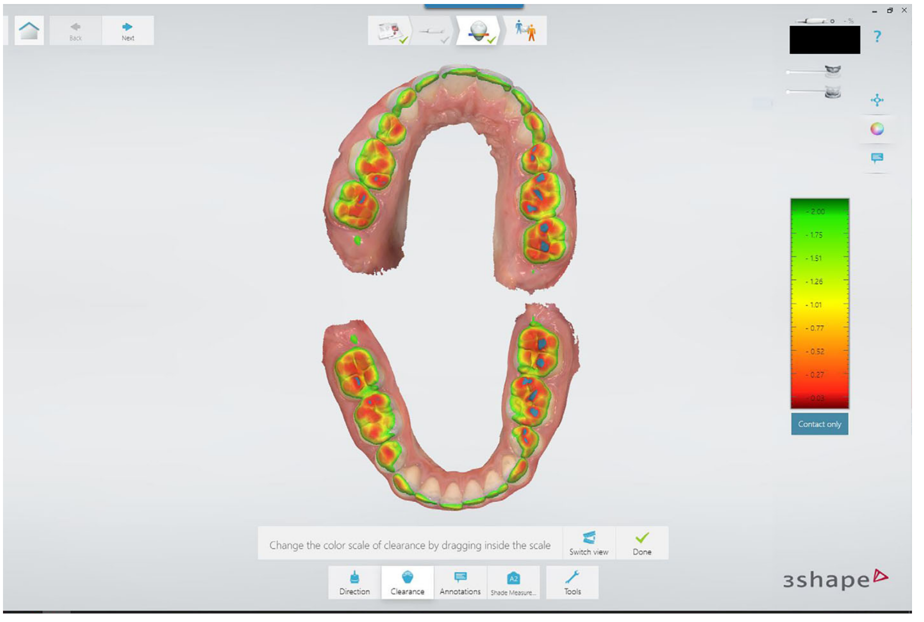

FIGURE 5 Occlusal contact area determination by using an IOSs (Trios 4, wireless; 3Shape A/S).

Future investigations may consider studying the reliability of the sensors during repeated use to assess the manufacturer's recommendation that a particular sensor be assigned to a specific patient for the purposes of follow-up.[90–93] Additionally, additional studies are needed for further analyze different aspects of the system such as response times, sensor hysteresis and saturation, and the effect of performing a preloading process.[90–93]

5 | CONCLUSIONS

- Digital occlusion technologies including IOSs, jaw tracking systems, and computerized occlusal analysis devices provide powerful diagnostic and designing tools for prosthodontic care. However, the accuracy of these digital technologies for acquiring and analyzing the static and dynamic occlusion need to be further analyzed.
- The accuracy of the maxillomandibular relationship recorded by using IOSs can be influenced by different factors such as articulation technique, characteristics of the virtual occlusal records (number, dimension, and location), length of the digitized arches, presence of edentulous areas (position and size), and IOS technology. The majority of the reviewed studies only analyzed the accuracy of the maxillomandibular relationship at MIP. Additionally, mesh interpenetrations or occlusal collisions that can affect the accuracy of the recorded maxillomandibular relationship need to be further studied.
- Different commercially and non-commercially available digital jaw systems have been described including ultrasonic systems, infrared optical camera systems, photometric devices, IOSs, and AI algorithms, but their accuracy remains unclear. Studies are needed to assess the accuracy, efficiency, and clinical impact when these recorded mandibular motions are used for designing interim and definitive restorations or occlusal devices.
- Computerized occlusal analysis systems provide a reliable method for measuring the OCA at maximum bite force; however, conventional occlusal registration methods seem to provide the highest reliability and validity technique for measuring OCA.

DISCLOSURE

The authors declare that they do not have any financial interest in the companies whose materials are included in this article.

FUNDING INFORMATION

This research did not receive any specific grant from funding agencies in the public, commercial, or non-profit sectors.

DATA AVAILABILITY STATEMENT

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

ORCID

Marta Revilla-León https://orcid.org/0000-0003-2854-1135

REFERENCES

1. The glossary of prosthodontic terms: ninth edition. *J Prosthet Dent*. 2017;117:1-5.
2. Luce CE. The movements of the lower jaw. *Boston Med Surg J*. 1889; 121:8-11.
3. Walker WE. Movements of the mandibular condyles and dental articulation. *Dent Cosmos*. 1896;38:573-583.
4. Hildebrand GY. Studies in the masticatory movements of the human lower jaw. *Arch Physiol*. 1931;61(suppl):61.
5. Hall RE. An analysis of the work and ideas of investigators and authors of relations and movements of the mandible. *J Am Dent Assoc*. 1922;16:1642-1693.
6. Posselt U. Studies in the mobility of the human mandible. *Acta Odontol Scand*. 1952;10:19-160.
7. Shanahan TEJ, Leff A. Mandibular and articulator movements. *J Prosthetic Dent*. 1959;9:941-945.
8. Swanson KH. A new method of recording gnathological movements. *Northwest Dent*. 1966;45:99-101.
9. Lee RL. Jaw movements engraved in solid plastic for articulator controls. *J Prosthet Dent*. 1969;22:209-224.
10. Gibbs CH, Messerman T, Reswick JB, Derda HJ. Functional movements of the mandible. *J Prosthet Dent*. 1971;26:604-620.
11. Gibbs CH, Messerman T. *The Case Gnathic Replicator for the Investigation of Mandibular Movements*. Case Institute of Technology, Engineering Design Center; 1966.
12. Jankelson B, Swain CW, Crane PF, Radke JC. Kinesiometric instrumentation: a new technology. *J Am Dent Assoc*. 1975;90(4):834-840.
13. Slavicek R. Clinical and instrumental functional analysis for diagnosis and treatment planning. Part 5. Axiography. *J Clin Orthod*. 1988; 22(10):656-667.
14. Klatsky M. A cinefluorographic study of the human masticatory apparatus in function. *Am J Orthod and Oral Surg*. 1940;26:664-670.
15. Jankelson B, Hoffman GM, Hendron JA. The physiology of the stomatognathic system. *Jada*. 1953;46:375-386.
16. Yurkstas AA, Emerson WH. A study of tooth contact during mastication with artificial dentures. *J Prosthet Dent*. 1954;4:168-174.
17. Lewin A, van Rensburg LB, Lemmer J. A method of recording the movement of a point on the jaw. *J Dent Assoc S Afr*. 1974;29: 395-397.
18. Karlsson S. Recording of mandibular movements by intraorally placed light emitting diodes. *Acta Odontol Scand*. 1977;35:111-117.
19. Röhrle O, Waddell JN, Foster KD, et al. Using a motion-capture system to record dynamic articulation for application in CAD/CAM software. *J Prosthodont*. 2009;18:703-710.
20. Mitchell DL, Wilkie ND. Articulators through the years. Part I. Up to 1940. *J Prosthet Dent*. 1978;39:330-338.
21. Weinberg LA. An evaluation of basic articulators and their concepts: part I. Basic concepts. *J Prosthet Dent*. 1963;13:622-644.
22. Weinberg LA. An evaluation of basic articulators and their concepts: part II. Arbitrary, positional, semi adjustable articulators. *J Prosthet Dent*. 1963;13:645-663.
23. Weinberg LA. An evaluation of basic articulators and their concepts: part III. Fully adjustable articulators. *J Prosthet Dent*. 1963;13: 873-888.
24. Dixon DL. Overview of articulation materials and methods for the prosthodontic patient. *J Prosthet Dent*. 2000;83:235-247.
25. Nishigawa K, Satsuma T, Shigemoto S, Bando E, Nakano M, Ishida O. Development of a novel articulator that reproduced jaw movement with six-degree-of-freedom. *Med Eng Phys*. 2007;29(5): 615-619.
26. Enciso R, Memon A, Fidaleo DA, et al. The virtual craniofacial patient: 3D jaw modeling and animation. *Stud Health Technol Inform*. 2003;94: 65e-71e.
27. Hanssen N, Ruge S, Kordass B. SICAT function: anatomical real-dynamic articulation by merging cone beam computed tomography and jaw motion tracking data. *Int J Comput Dent*. 2014;17:65-74.
28. Sójka A, Huber J, Kaczmarek E, Hędzelek W. Evaluation of mandibular movement functions using instrumental ultrasound system. *J Prosthodont*. 2017;26:123-128.
29. Kwon JH, Im S, Chang M, Kim JE, Shim JS. A digital approach to dynamic jaw tracking using a target tracking system and a structured-light three- dimensional scanner. *J Prosthodont Res*. 2019;63: 115-119.
30. Bapelle M, Dubromez J, Savoldelli C, et al. Modjaw® device: analysis of mandibular kinematics recorded for a group of asymptomatic subjects. *Cranio*. 2021;1-7.
31. Li W, Li L, Wang Y, et al. Accuracy of recording edentulous jaw relations by using an optical jaw tracking system: an in vitro study. *Int J Prosthodont*. 2022;35:302-310.
32. Revilla-León M, Zeitler JM, Kois JC. Digital maxillomandibular relationship and mandibular motion recording by using an optical jaw tracking system to acquire a dynamic virtual patient. *J Prosthet Dent*. Forthcoming. 2022.
33. Bedrossian EA, Bedrossian E, Kois JC, Revilla-León M. Use of an optical jaw-tracking system to record mandibular motion for treatment planning and designing interim and definitive prostheses: a dental technique. *J Prosthet Dent*. Forthcoming. 2022.
34. Tian SK, Dai N, Li LL, Li W, Sun Y, Cheng X. Three-dimensional mandibular motion trajectory-tracking system based on BP neural network. *Math Biosci Eng*. 2020;17(5):5709-5726.
35. Zoss G, Beeler T, Gross M, Bradley D. Accurate markerless jaw tracking for facial performance capture. *ACM Trans Graph*. 2019; 38:1-8.
36. Xia JJ, Chang YB, Gateno J, Xiong Z, Zho X. Automated digital dental articulation. *Med Image Comput Comput Assist Interv*. 2010;13(Pt 3): 278-286.
37. Bando E, Nishigawa K, Nakano M, et al. Current status of researches on jaw movement and occlusion for clinical application. *Jpn Dent Sci Rev*. 2009;45:83-97.
38. Revilla-Leon M, Frazier K, da Costa JB, et al. Intraoral scanners: an American Dental Association Clinical Evaluators Panel survey. *J Am Dent Assoc*. 2021;152(8):669-670.e2.
39. Hennen MV, Blum H, Dammaschke T. Accuracy and reproducibility of the visualization of occlusal contact points using analog articulating foil or digital intraoral scanners in vitro. *Int J Comput Dent*. 2022; 25(2):173-180.
40. Wang Q, Zhao Z, Zhou M, et al. In vivo evaluation of the reliability and validity of three digital occlusion analysis methods. *J Dent*. 2022; 127:104355.
41. Ayuso-Montero R, Mariano-Hernandez Y, Khoury-Ribas L, Rovira-Lastra B, Willaert E, Martinez-Gomis J. Reliability and validity of t-scan and 3d intraoral scanning for measuring the occlusal contact area. *J Prosthodont*. 2020;29(1):19-25.
42. Abdulateef S, Edher F, Hannam AG, Tobias DL, Wyatt CCL. Clinical accuracy and reproducibility of virtual interocclusal records. *J Prosthet Dent*. 2020;124(6):667-673.

43. Edher F, Hannam AG, Tobias DL, Wyatt CCL. The accuracy of virtual interocclusal registration during intraoral scanning. *J Prosthet Dent*. 2018;120(6):904-912.
44. Delong R, Ko CC, Anderson GC, et al. Comparing maximum intercuspal contacts of virtual dental patients and mounted dental casts. *J Prosthet Dent*. 2002;88(6):622-630.
45. Maness WL. Comparison of the duration of occlusal contacts during habitual closure using the digital occlusal sensor. *J Dent Res*. 1986; 65:185.
46. Maness WL, Chapman R, Dario LD. Laboratory evaluation of a direct reading digital occlusal sensor. *J Prosthet Dent*. 1985;54:744.
47. Hsu ML, Palla S, Gallo LM. Sensitivity and reliability of the t-scan system for occlusal analysis. *J Craniomandib Disord*. 1992;6:17-23.
48. Lee W, Kwon HB, Kim MJ, Lim YJ. Determination of the reliability and repeatability of a quantitative occlusal analyzer by using a piezoelectric film sensor: an in vitro study. *J Prosthet Dent*. 2022;127(2): 331-337.
49. Türp JC, Greene CS, Strub JR. Dental occlusion: a critical reflection on past, present and future concepts. *J Oral Rehabil*. 2008;35:446e-453e.
50. McNeill C. *Science and Practice of Occlusion*. Quintessence; 1997: 306e-322e.
51. Shafagh I, Yoder JL, Thayer KE. Diurnal variance of centric relation position. *J Prosthet Dent*. 1975;34(5):574-582.
52. Tripodakis AP, Smulow JB, Mehta NR, Clark RE. Clinical study of location and reproducibility of three mandibular positions in relation to body posture and muscle function. *J Prosthet Dent*. 1995;73(2): 190-198.
53. Campos AA, Nathanson D, Rose L. Reproducibility and condylar position of a physiologic maxillomandibular centric relation in upright and supine body position. *J Prosthet Dent*. 1996;76(3):282-287.
54. Coelho MF, Cavalcanti B, Claro Neves AC, et al. Influence of dental chair backrest inclination on the registration of the mandibular position. *J Prosthet Dent*. 2015;114(5):693-695.
55. Nakamura K, Minami I, Wada J, Ikawa Y, Wakabayashi N. Head position affects the direction of occlusal force during tapping movement. *J Oral Rehabil*. 2018;45(5):363-370.
56. Wiechens B, Brockmeyer P, Wassmann T, Rödiger M, Wiessner A, Bürgers R. Time of day-dependent deviations in dynamic and static occlusion: a prospective clinical study. *J Prosthet Dent*. Forthcoming. 2022.
57. Schelb E, Kaiser DA, Brukl CE. Thickness and marking characteristics of occlusal registration strips. *J Prosthet Dent*. 1985;54:122e-126e.
58. Anderson GC, Schulte JK, Aeppli DM. Reliability of the evaluation of occlusal contacts in the intercuspal position. *J Prosthet Dent*. 1993;70: 320e-323e.
59. Gazit E, Fitzig S, Lieberman MA. Reproducibility of occlusal marking techniques. *J Prosthet Dent*. 1986;55:505-509.
60. Pröschel PA. Chewing patterns in subjects with normal occlusion and with malocclusions. *Semin Orthod*. 2006;12:138-149.
61. Sharma A, Rahul GR, Poduval ST, Shetty K, Gupta B, Rajora V. History of materials used for recording static and dynamic occlusal contact marks: a literature review. *J Clin Exp Dent*. 2013;5:48-53.
62. Morsy N, El Kateb M. Accuracy of intraoral scanners for static virtual articulation: a systematic review and meta-analysis of multiple outcomes. *J Prosthet Dent*. Forthcoming. 2022.
63. Shadid R, Sadaqah N. Accuracy of virtual static articulation: a systematic review. *Int J Prosthodont*. 2022;35(5):627-646.
64. Revilla-León M, Kois DE, Kois JC. A guide for maximizing the accuracy of intraoral digital scans. Part 1: operator factors. *J Esthet Restor Dent*. Forthcoming. 2022;35(1):230-240.
65. Revilla-León M, Kois DE, Kois JC. A guide for maximizing the accuracy of intraoral digital scans. Part 2: patient factors. *J Esthet Restor Dent*. Forthcoming. 2022;35(1):241-249.
66. Revilla-León M, Gohil A, Barmak AB, et al. Influence of ambient temperature changes on intraoral scanning accuracy. *J Prosthet Dent*. Forthcoming. 2022.
67. Revilla-León M, Jiang P, Sadeghpour M, et al. Intraoral digital scans-part 1: influence of ambient scanning light conditions on the accuracy (trueness and precision) of different intraoral scanners. *J Prosthet Dent*. 2020;124(3):372-378.
68. Revilla-León M, Jiang P, Sadeghpour M, et al. Intraoral digital scans: part 2-influence of ambient scanning light conditions on the mesh quality of different intraoral scanners. *J Prosthet Dent*. 2020;124(5): 575-580.
69. Kong L, Li Y, Liu Z. Digital versus conventional full-arch impressions in linear and 3D accuracy: a systematic review and meta-analysis of in vivo studies. *Clin Oral Investig*. 2022;26:5625-5642.
70. Aragón ML, Pontes LF, Bichara LM, et al. Validity and reliability of intraoral scanners compared to conventional gypsum models measurements: a systematic review. *Eur J Orthod*. 2016;38:429-434.
71. Kernen F, Schlager S, Seidel Alvarez V, et al. Accuracy of intraoral scans: an in vivo study of different scanning devices. *J Prosthet Dent*. 2021;128:1303-1309.
72. Revilla-León M, Subramanian SG, Özcan M, Krishnamurthy VR. Clinical study of the influence of ambient light scanning conditions on the accuracy (trueness and precision) of an intraoral scanner. *J Prosthodont*. 2020;29:107-113.
73. Braian M, Wennerberg A. Trueness and precision of 5 intraoral scanners for scanning edentulous and dentate complete-arch mandibular casts: a comparative in vitro study. *J Prosthet Dent*. 2019;122(2): 129-136.
74. Revilla-León M, Alonso Pérez-Barquero J, Zubizarreta-Macho Á, et al. Influence of the number of teeth and location of the virtual occlusal record on the accuracy of the maxillo-mandibular relationship obtained by using an intraoral scanner. *J Prosthodont*. Forthcoming. 2022;32(3):253-258.
75. Revilla-León M, Gómez-Polo M, Zeitler JM, Barmak AB, Kois JC, Pérez-Barquero JA. Does the available interocclusal space influence the accuracy of the maxillomandibular relationship captured with an intraoral scanner? *J Prosthet Dent*. Forthcoming. 2022.
76. Osnes C, Wu J, Ferrari M, Joda T, Keeling A. Sources of error in maximum intercuspation from complete dentate full-arch intraoral scans in vitro. *Int J Comput Dent*. 2021;24(3):283-291.
77. Chang YB, Xia JJ, Gateno J, Xiong Z, Zhou X, Wong STC. An automatic and robust algorithm of reestablishment of digital dental occlusion. *IEEE Trans Med Imaging*. 2010;29(9):1652-1663.
78. Chang YB, Xia JJ, Gateno J, et al. In vitro evaluation of new approach to digital dental model articulation. *J Oral Maxillofac Surg*. 2012;70(4): 952-962.
79. Stavness IK, Hannam AG, Tobias DL, Zhang X. Simulation of dental collisions and occlusal dynamics in the virtual environment. *J Oral Rehabil*. 2016;43(4):269-278.
80. Revilla-León M, Gómez-Polo M, Vyas S, et al. Artificial intelligence applications in restorative dentistry: a systematic review. *J Prosthet Dent*. 2022;128(5):867-875.
81. Revilla-León M, Gómez-Polo M, Vyas S, et al. Artificial intelligence models for tooth-supported fixed and removable prosthodontics: a systematic review. *J Prosthet Dent*. Forthcoming. 2021;129(2):276-292.
82. Revilla-León M, Gómez-Polo M, Vyas S, et al. Artificial intelligence applications in implant dentistry: a systematic review. *J Prosthet Dent*. Forthcoming. 2021;129(2):293-300.
83. Feng Y, Max L. Accuracy and precision of a custom camera-based system for 2-D and 3-D motion tracking during speech and nonspeech motor tasks. *J Speech Lang Hear Res*. 2014;57:426-438.
84. Farook TH, Rashid F, Alam MK, Dudley J. Variables influencing the device-dependent approaches in digitally analysing jaw movement-a systematic review. *Clin Oral Investig*. 2022;27:489-504.

85. Woodford SC, Robinson DL, Mehl A, Lee PVS, Ackland DC. Measurement of normal and pathological mandibular and temporomandibular joint kinematics: a systematic review. *J Biomech*. 2020;111:109994.
86. Revilla-León M, Fernández-Estevan L, Barmak AB, et al. Accuracy of the maxillomandibular relationship at centric relation position recorded by using 3 different intraoral scanners with or without an optical jaw tracking system: an in vivo pilot study. *J Dent*. 2023;104478.
87. Halperin GC, Halperin AR, Norling BK. Thickness, strength, and plastic deformation of occlusal registration strips. *J Prosthet Dent*. 1982;48(5):575-578.
88. Luo Q, Ding Q, Zhang L, Xie Q. Analyzing the occlusion variation of single posterior implant-supported fixed prostheses by using the T-scan system: a prospective 3-year follow-up study. *J Prosthet Dent*. 2020;123(1):79-84.
89. Helms RB, Katona TR, Eckert GJ. Do occlusal contact detection products alter the occlusion? *J Oral Rehabil*. 2012;39(5):357-363.
90. Cerna M, Ferreira R, Zaror C, Navarro P, Sandoval P. Validity and reliability of the T-Scan(®) III for measuring force under laboratory conditions. *J Oral Rehabil*. 2015;42(7):544-551.
91. Jeong MY, Lim YJ, Kim MJ, Kwon HB. Comparison of two computerized occlusal analysis systems for indicating occlusal contacts. *J Adv Prosthodont*. 2020;12(2):49-54.
92. Dias RAB, Rodrigues MJP, Messias AL, Guerra FADA, Manfredini D. Comparison between conventional and computerised methods in the assessment of an occlusal scheme. *J Oral Rehabil*. 2020;47(2):221-228.
93. Mitchem JA, Katona TR, Moser EAS. Does the presence of an occlusal indicator product affect the contact forces between full dentitions? *J Oral Rehabil*. 2017;44(10):791-799.

How to cite this article: Revilla-León M, Kois DE, Zeitler JM, Att W, Kois JC. An overview of the digital occlusion technologies: Intraoral scanners, jaw tracking systems, and computerized occlusal analysis devices. *J Esthet Restor Dent*. 2023;1-10. doi:10.1111/jerd.13044